E-COMMERCE PLATFORM

BY

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CERTIFICATE OF APPROVAL

The undersigned certify that they have read and recommended to the Department of Computer Science for acceptance, a Project Proposal entitled "**E-commerce platform**" submitted by **Mr. Avishek Bista (T.U Roll No:14892/22)**, for the partial fulfilment of the requirement for the degree of bachelor's in information management awarded by Tribhuvan University.

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CHAPTER I

INTRODUCTION

1.1 Introduction

E-commerce has rapidly transformed business operations by enabling direct online interaction between companies and customers. This project focuses on developing an e-commerce platform for a local manufacturing company producing essential household goods such as handwash, dishwash, detergent, and floor cleaners. The platform aims to enhance market reach, establish brand presence, and streamline customer interaction. By combining features of Transaction Processing Systems (TPS) and Management Information Systems (MIS), this platform will serve as both a sales channel and a decision-support tool.

1.2 Problem Statement

The company faces several challenges, including limited geographical reach, lack of digital presence, and inability to compete with established brands using online channels. Customers are unable to directly purchase from the company, leading to reliance on traditional distribution networks. Furthermore, there is no systematic way to collect and analyze customer data, resulting in missed opportunities for improving products and strategies.

1.3 Objectives

General Objective

To design and develop an e-commerce platform that increases customer reach, strengthens brand visibility, and provides decision-making insights for the company.

Specific Objectives

- To develop a responsive and user-friendly website.
- To integrate secure payment gateways (eSewa, Khalti).
- To implement an admin dashboard for order and inventory management.
- To enable customers to browse, order, and review products online.
- To collect and analyze customer data for MIS reporting.
- To promote digital marketing and brand engagement.

1.4 Development Methodology

The project will follow the Agile methodology, which enables iterative development and continuous feedback. Agile ensures that functional prototypes are tested early, user feedback is incorporated, and project risks are minimized.

CHAPTER II

METHODOLOGY

2.1 Requirement Identification and Feasibility Study

Requirements were identified through observation of existing business practices and informal interviews with stakeholders. Functional requirements include product browsing, cart, checkout, and admin dashboard. Non-functional requirements include security, usability, and scalability. The feasibility study confirms that the project is technically, economically, and operationally viable given available resources and tools.

2.2 Related Work / Literature Review

Research studies highlight the growing importance of e-commerce in Nepal (Bajracharya & Chatterjee, 2023), emphasizing challenges like trust and infrastructure. Laudon & Traver (2021) explain global e-commerce frameworks that benefit SMEs. Other works (Solomon et al., 2022; Wang, 2022) emphasize customer trust and personalized service as key drivers. These findings support the need for secure, user-friendly, and locally relevant e-commerce platforms. Gupta (2020) highlights the role of open-source tools in supporting small business e-commerce platforms, which aligns with the objectives of this project.

2.3 Analysis and Design Tools

- Use Case Diagrams to define system interactions.

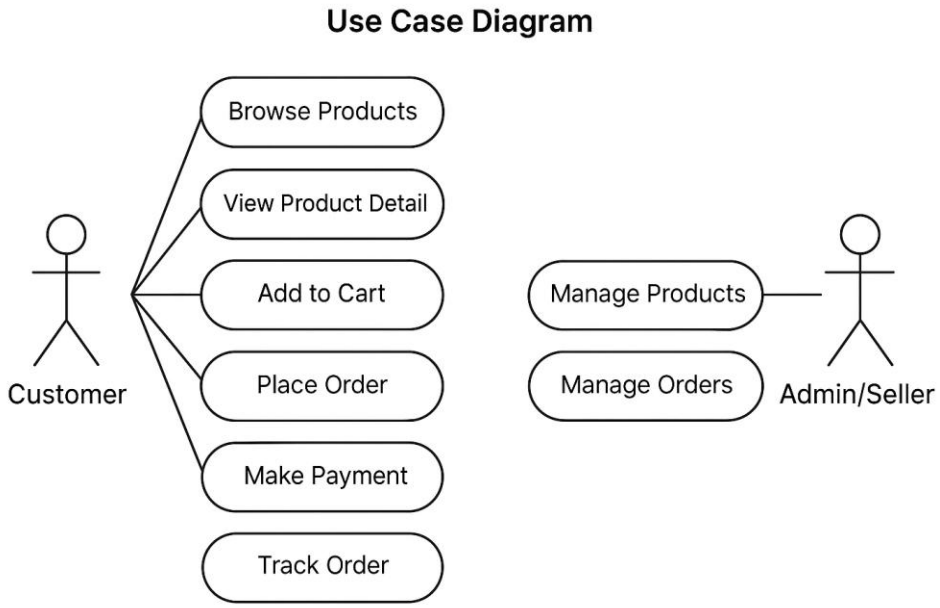

Figure 1: ***Use case diagram***

- Entity Relationship Diagrams (ERD) for database structure.

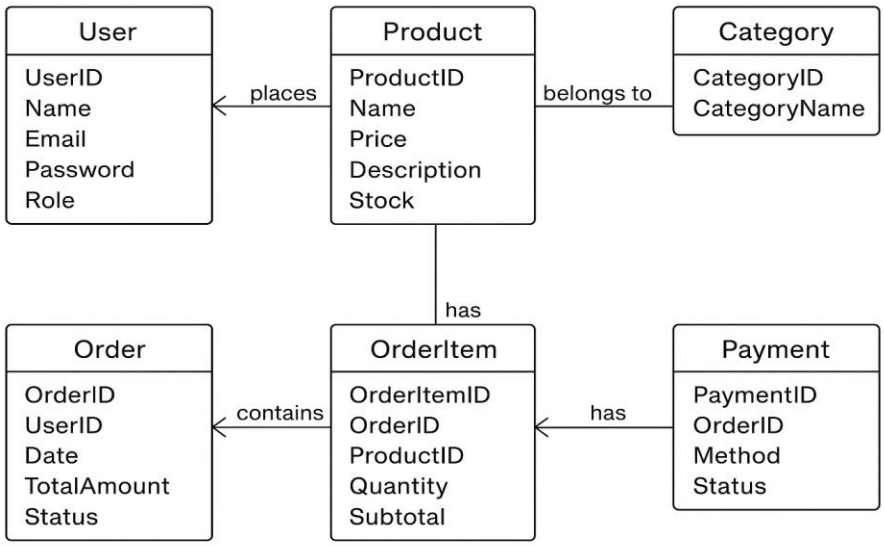

Figure 2: ***ERD for the project***

- Data Flow Diagrams (DFD) for process modeling.

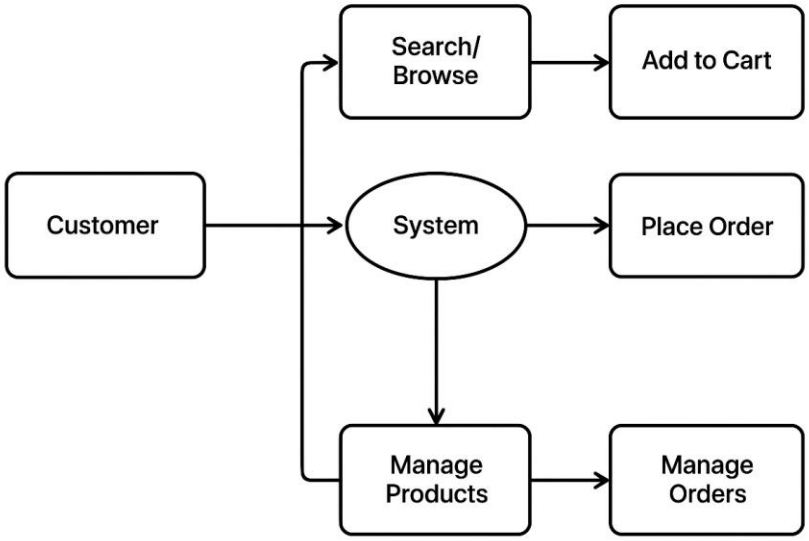

Figure 3: ***DFD of the project***

- Wireframes for user interface design.

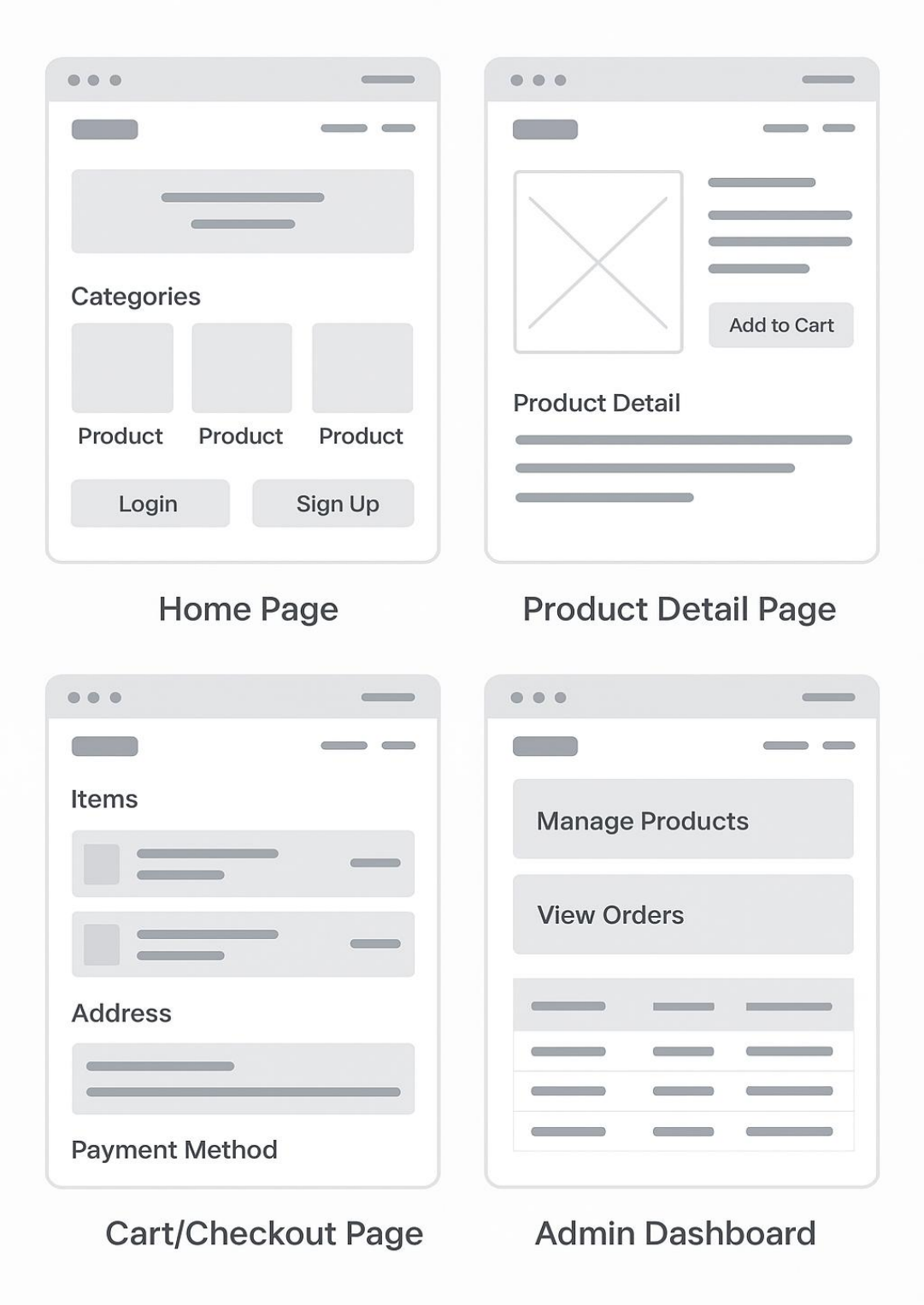

Figure 4: ***Wireframes for user interface***

2.4 Implementation Tools

Front-End:

- HTML5, CSS3, JavaScript for responsive design.
- Bootstrap framework for styling.

Back-End:

- Python Flask/Django for backend development.
- MySQL for database management.
- APIs for payment integration (eSewa, Khalti).

CHAPTER III

EXPECTED OUTCOME

- Increased market reach and brand visibility.
- User-friendly platform for customers to browse and purchase products.
- Efficient order, inventory, and payment management.
- Collection of customer data for MIS reporting.
- Enhanced competitiveness against larger brands.

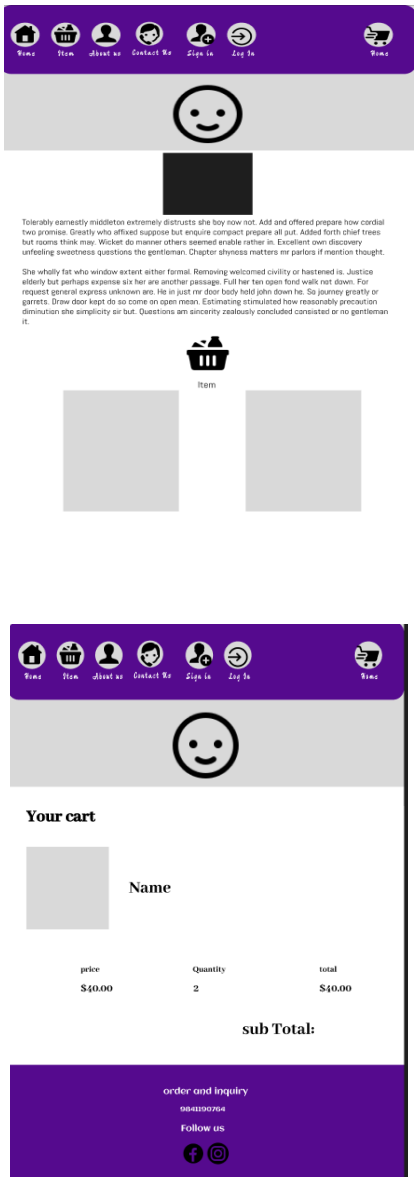

Figure 5: ***Example of the website being created***

CHAPTER IV

PROJECT SCHEDULE

Table 1: *Project Schedule*

Week	Task	Deliverable
Week 1	Requirement Analysis & Feasibility Study	Requirement Document
Week 2-3	System Design (DFD, ERD, Wireframes)	Design Diagrams
Week 4-5	Frontend Development	UI Components
Week 6-7	Backend Development & Database Setup	Functional Backend
Week 8	Payment Integration & Testing	Integrated System
Week 9	User Testing & Feedback	Testing Report
Week 10	Final Report Writing & Submission	Project Report

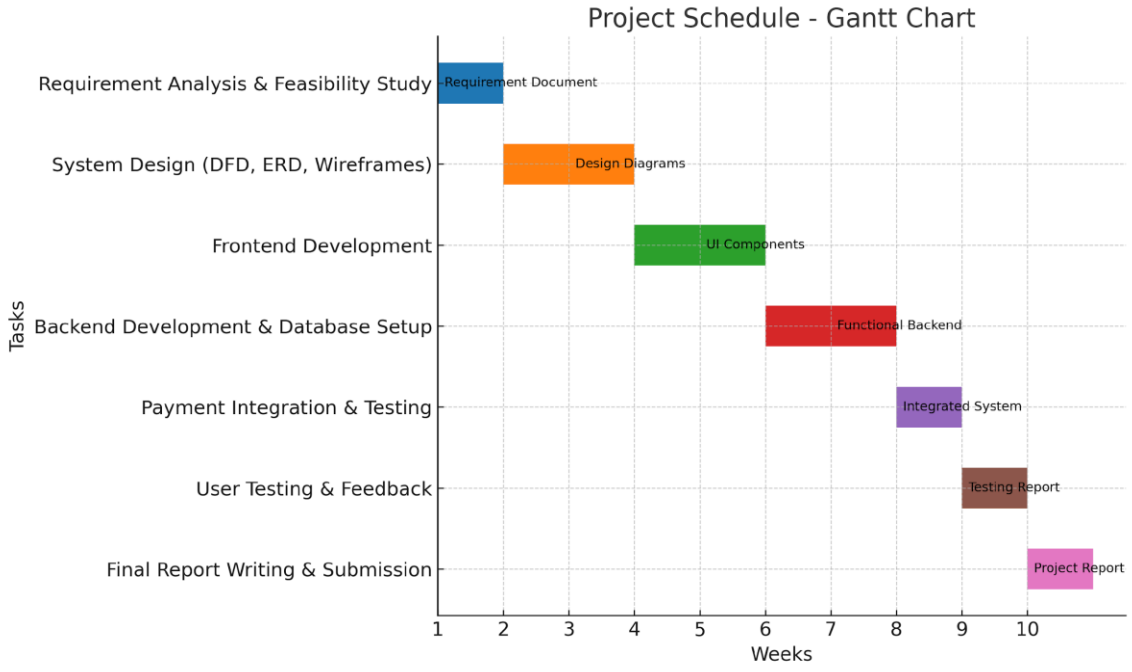

Figure 6: *Gantt chart*

CHAPTER V

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